

When the Algorithm Meets the Absent Facility: The AI–Physical Execution Gap in India’s E-Waste Circular Economy

Aditya Sharma¹, Hossein Tabasi^{1*}, Anurag Rana¹, Pankaj Vaidya¹

¹ Shoolini University of Biotechnology and Management Sciences, Solan, Himachal Pradesh 173229, India

* Corresponding author: Hossein Tabasi — hosseintabasi@shooliniuniversity.com

Abstract

Recommendation models can now classify an end-of-life electronic device, score its condition, price its residual value, and rank circular recovery options in well under a second. That computational cheapness has outrun the physical system in which the recommendations are supposed to land. India is among the world’s three largest sources of electronic waste. The Global E-waste Monitor 2024 records 4.1 million tonnes generated in India in 2022, third after China (Baldé et al., 2024). A NITI Aayog synthesis of the same international accounting family puts generation at about 6.19 million tonnes in 2024 and on course for 14 million tonnes by 2030 (NITI Aayog, 2026). India’s own official series, estimated by the CPCB from producer sales and prescribed equipment lives, is much smaller: 1,254,286.55 tonnes in FY 2023–24 and 1,397,955.59 tonnes in FY 2024–25, with formal collection-and-processing shares of 61.94 percent and 70.71 percent against those official denominators (MoEFCC, 2025). Against any of these volumes the licensed layer is thin. As of 9 February 2025 the CPCB portal listed 322 registered recyclers (reported capacity 2,208,918.064 tonnes a year) and 72 registered refurbishers (92,042.18 tonnes a year) (Press Information Bureau, 2025). Independent industry estimates still assign about 90 percent of collection and 70 percent of recycling to informal actors (ICEA, 2024).

The claim of this chapter is straightforward and easy to miss: an AI recommendation is not a circular outcome. Using India’s semiconductor-dense device stream, and smartphones in particular, as the empirical setting, we define the **AI–Physical Execution Gap (AIPEG)** as the systematic divergence between the circular pathway an algorithm ranks as optimal and the subset of pathways that a territory’s physical, institutional, logistical, and market infrastructure can actually deliver. Executability is specified as five simultaneous conditions—**availability, accessibility, capacity, viability, and legibility**. From those conditions we construct an Execution Feasibility Index and a Circularity Execution Ratio, locate the binding constraint at each stage of the smartphone circular chain, and formalise the reverse-logistics economics of the rural–urban divide. India’s Extended Producer Responsibility (EPR) rules and digital public infrastructure are then examined as possible gap-narrowing or gap-widening designs.

A reproducible computation under the chapter’s stated parameter assumptions, and a 30-origin Indian geographic panel, shows that for small devices the consolidation term dominates haulage by more than two orders of magnitude (spread ₹248 versus ₹0.56 per device). Harvesting remains at $\phi = 0$ wherever no licence exists. On the geographic panel the Circularity Execution Ratio for hinterland origins is 0 without an aggregation node and 1 once a node is introduced—provided a licensed refurbishment pathway exists. The chapter therefore proposes **Infrastructure-Constrained Circular AI**: a design stance in which facility availability, distance, transport economics, and real-time capacity enter the objective function as constraints, not as friction discovered after the recommendation has already been issued.

Keywords: circular economy; artificial intelligence; electronic waste; India; reverse logistics; Extended Producer Responsibility; infrastructure constraints; emerging markets; semiconductor devices; rural–urban divide

1. Introduction: The Recommendation That Cannot Be Executed

Take a five-year-old smartphone in a village in Nuapada district, Odisha. The screen is intact. Battery capacity has fallen to about 60 percent of rating. The system-on-chip, camera module, and radio-frequency front end still function. Fed with images, diagnostic telemetry, secondary-market prices, and a materials database, a competent model will return a tidy circular plan: harvest the display and camera for the spare-parts market, replace the battery, refurbish the handset, and sell it into a price-sensitive segment. It will also emit a recovery value, an avoided-emissions figure, and a confidence score—quickly, cheaply, and often with defensible accuracy.

None of that plan has a local executor. No registered refurbisher sits within several hundred kilometres. The block has no formal collection point. The nearest authorised dismantler is in a metropolitan cluster, and moving a single 180-gram device there costs an order of magnitude more than the value recoverable from it. The aggregator who could bundle the handset into a viable shipment is informal, has no digital footprint, and does not appear in the model's world. In practice the phone is sold for cash to a travelling scrap buyer, passed along an untraceable chain into an informal cluster, stripped of high-value components by hand, and the residual board fraction is processed in ways that recover only part of the available metal while shifting environmental and occupational harm onto workers and nearby communities.

The inference was not false. It was unused. That unused inference is the object of this chapter.

Scholarship on artificial intelligence and the circular economy has grown quickly, and much of it is optimistic. Documented applications include machine-vision sorting, predictive maintenance, digital product passports, residual-value estimation, demand forecasting for secondary materials, and platform-mediated reverse marketplaces (Ellen MacArthur Foundation, 2019; Ghoreishi & Happonen, 2020; Sarc et al., 2019; Wilts et al., 2016). The implicit theory of change is that better identification, classification, prediction, and optimisation will move materials through higher-value loops. That theory is plausible where facilities are dense, capitalised, and standardised enough to carry out most of what an optimiser proposes. India's physical system does not meet that precondition. Once that is granted, "better inference" ceases to be a sufficient theory of circular improvement.

The limiting factor in emerging-market circularity is no longer the ability to infer. It is the ability to execute. The two have come apart. We name that decoupling the **AI-Physical Execution Gap (AIPEG)**: the systematic, measurable difference between the set of circular pathways an algorithm identifies as optimal and the subset of those pathways that a given territory's physical, institutional, logistical, and market infrastructure can execute at acceptable cost and environmental impact. AIPEG is not a change-management problem sitting downstream of a good model. It is a structural property of the coupled system. It varies with geography and device class. Standard recommenders do not emit a signal when their output is ignored, so the gap can widen while reported performance still improves.

That invisibility has practical stakes. Programmes can post rising circularity on the strength of recommendations issued, EPR certificates traded, or platform-reported diversion even as material outcomes stall or worsen. We call the resulting artefact **phantom circularity**: algorithmically legitimated circular intent that is never physically realised, yet is counted, reported, and, under certificate-based compliance, monetised.

India is a demanding test of this claim. It is among the three largest generators of e-waste (Baldé et al., 2024), and generation is growing faster than the infrastructure meant to absorb it (NITI Aayog, 2026). Formal recovery capacity sits in a small number of industrial clusters, while India's population of about 1.44 billion remains majority rural in settlement structure (United Nations DESA, 2024) and 523.58 million wireless subscriptions were rural as of 31 December 2024 out of 1,150.66 million wireless subscriptions nationally (TRAI, 2025). At the same time, the E-Waste (Management) Rules, 2022, the CPCB EPR portal, and India's digital public infrastructure together form one of the more digitally mediated waste-governance regimes among

emerging economies—hence a real test of whether digital coordination can stand in for physical capacity. Composition matters as well. The stream is increasingly made of small, semiconductor-dense, high-value-per-kilogram devices, principally smartphones, whose circular economics are unusually sensitive to logistics cost, the variable current recommendation systems are least likely to represent.

1.1 Research questions

Three questions organise the chapter.

- **RQ1.** How should the divergence between the circular pathway an unconstrained optimiser recommends for an Indian end-of-life smartphone and the pathway its surrounding infrastructure can actually execute be defined and measured?
- **RQ2.** Which of the executability conditions binds first, and does the binding condition change systematically with distance from a formal facility cluster?
- **RQ3.** Do India's certificate-based Extended Producer Responsibility regime and its digital public infrastructure stack narrow the execution gap, or does the design of either widen it?

Sections 2 and 3 answer RQ1 conceptually and formally. Sections 4 to 6 address RQ2 through India's material substrate, a stage-by-stage bottleneck analysis, a break-even computation, and a reproducible geographic panel (Section 6.5). Sections 7 and 8 take up RQ3 and the design consequences. The code and datasets are described in Appendix C.

2. Conceptual Foundations: From Circular Ambition to Circular Capability

2.1 The R-hierarchy and its silent assumption

Circular-economy practice is commonly organised as a hierarchy of retention strategies—refuse, rethink, reduce, reuse, repair, refurbish, remanufacture, repurpose, recycle, recover—ranked by how much product function and embodied value each preserves (Kirchherr et al., 2017; Potting et al., 2017; Reike et al., 2018). The ranking is normative. Higher R-strategies are preferred because they keep more of the energy, labour, and material complexity already locked in the artefact. Geissdoerfer et al. (2017) and Blomsma and Brennan (2017) describe how previously scattered ideas hardened into a governing paradigm; Ghisellini et al. (2016) review its spread across policy regimes.

The hierarchy is silent on who performs each R, where, at what cost, and at what throughput. It ranks desirability and treats availability as given. In dense industrial economies that assumption is often close enough that it never becomes the analytical problem. In India it is wrong in a specific way: over most of the national territory the higher R-strategies have no local executor. A ten-item menu then collapses, in practice, to informal recycling or landfill—regardless of what the artefact would merit and regardless of what an algorithm ranks first.

Velenturf and Purnell (2021) argue that circular-economy research has attended more to material flows than to the institutional and infrastructural systems that carry those flows. Corvellec et al. (2022) go further, treating circular-economy discourse as a device that can displace attention from material constraints on implementation. The extension we offer is constructive rather than dismissive. Circularity is not a fiction. The field has not named executability as an explicit variable, and in emerging markets that omission is not a second-order correction. It is often the dominant term.

2.2 What AI actually does in the circular economy

AI applications in circular systems cluster into four functional families. Each family stands in a different relation to physical infrastructure.

Identification and classification. Convolutional and transformer-based vision models classify device type, brand, model, and cosmetic condition from images; spectroscopic and X-ray systems classify material fractions on sorting lines (Gundupalli et al., 2017; Nowakowski & Pamuła, 2020; Sarc et al., 2019). These systems are substrate-coupled. They run inside a facility and are useful because the facility exists.

Prediction and forecasting. Models estimate remaining useful life, residual market value, component-failure probability, and secondary-material demand. Prediction is substrate-independent. It can be run anywhere, on any device, at negligible cost, and accuracy does not require a nearby plant.

Routing and optimisation. Systems assign end-of-life devices to R-pathways, plan collection routes, allocate volumes to facilities, and price transactions. This family is nominally substrate-aware. In the dominant literature and in commercial practice, however, the facility network is treated as an exogenous given: usually complete, always reachable.

Coordination. Platforms, digital product passports, marketplaces, and traceability systems match supply and demand among participating actors (Jensen et al., 2023). Coordination is substrate-conditional. It can raise the productivity of capacity that already exists. It cannot invent capacity that does not.

The argument turns on the asymmetry among these families. Substrate-independent capabilities scale quickly and cheaply, and they have. Substrate-coupled and substrate-conditional capabilities are bounded by physical assets that scale slowly, need capital, and are unevenly placed. A model could, in principle, classify every smartphone retired in India this year. The formal system can process only a fraction of that volume, and only in a few places. The recommendation layer has scaled by orders of magnitude. The execution layer has not.

2.3 The missing variable: executability

A circular recommendation is therefore treated here as a triple: a device state, a proposed retention strategy, and a designated executing facility. The recommendation is executable if and only if five conditions hold at once.

- **Availability** — a facility exists within the reachable system that is technically equipped and licensed to perform the strategy.
- **Accessibility** — the device can physically reach that facility along an existing, lawful, and functioning logistics chain.
- **Capacity** — the facility has throughput available to accept the device within a window before residual value decays below threshold.
- **Viability** — recovered value net of collection, aggregation, transport, processing, and compliance cost is positive for every actor whose participation the chain requires.
- **Legibility** — the transaction is recordable, verifiable, and creditable within the governing institutional framework, so that the actors who perform it are recognised and compensated.

Failure of any one condition invalidates the recommendation, however accurate the underlying inference. Executability is conjunctive, not compensatory: strength on four dimensions does not offset failure on the fifth. That structure is why AIPEG is severe rather than merely inconvenient, and why further gains in model accuracy yield near-zero gains in circular outcomes once any single condition binds.

3. Defining the AI–Physical Execution Gap

3.1 Conceptual definition

The **AI–Physical Execution Gap** is the systematic divergence between the set of circular pathways an algorithmic system identifies as optimal for a population of end-of-life devices and the subset of those pathways that the surrounding physical, institutional, logistical, and market infrastructure can execute at acceptable cost and environmental impact within the devices’ value-retention window.

Four properties follow from the definition. First, the gap is systematic rather than stochastic. Geography, device class, and institutional design structure it, which makes it predictable and, in principle, addressable. Second, it is relational: a property of the pairing of an algorithm with a territory, not of either alone. The same model can show a small gap in a metropolitan cluster and a near-total gap two hundred kilometres away. Third, it is time-bounded. Residual value decays, so a pathway that becomes executable only after the value-retention window has closed was never executable in any operational sense. Fourth, it is invisible by default. Ordinary recommendation architectures do not flag unused output, so reported performance can improve while the gap widens without bound.

3.2 Five dimensions of the gap

The five executability conditions map onto five dimensions of the gap. Each has characteristic causes, observables, and remediation instruments (Table 1).

Table 1. The five dimensions of the AI–Physical Execution Gap.

Dimension	Failure condition	Primary observable indicator	Dominant cause in India	Remediation instrument
Availability	No licensed facility can perform the strategy anywhere in the reachable system	Facility counts per device-population unit; strategy coverage ratio	Extreme scarcity of registered refurbishers relative to recyclers; near-absence of component-level recovery licensing	Licensing reform; capacity investment; strategy-specific authorisation categories
Accessibility	A facility exists but the device cannot lawfully or economically reach it	Mean and 90th-percentile distance to nearest capable facility; collection-point density	Spatial concentration of capacity in industrial clusters; absent rural collection infrastructure	Aggregation nodes; hub-and-spoke reverse logistics; postal and retail network integration
Capacity	The facility exists and is reachable but cannot accept the volume in time	Facility utilisation rate; queue time versus value-decay rate	Thin installed throughput; seasonal and campaign-driven volume spikes; no real-time capacity signalling	Capacity registers; dynamic allocation; buffer storage nodes
Viability	Execution is physically possible but value-negative for at least one required actor	Net margin per device by pathway and origin; break-even distance and minimum viable consignment	High fixed consignment cost for small devices; low formal buy-back prices versus informal cash offers	Floor prices; EPR fee modulation; origin-indexed subsidy; consolidation economics
Legibility	Execution occurs but cannot be recorded, verified, or credited	Share of flow with digital provenance; informal-sector integration rate	Structural exclusion of informal aggregators from the EPR portal architecture	Tiered registration; aggregator digital identity; verified handover protocols

The dimensions bind in chain order and combine multiplicatively rather than additively. Better logistics does not repair an availability failure. Extra plant capacity does not help a device that cannot reach the plant. Information does not repair a viability failure. The measurement approach below encodes that multiplicative structure. It also helps explain a recurring policy puzzle: large outlays along one dimension often produce little aggregate change, because another dimension remains binding.

3.3 Operationalisation: the Execution Feasibility Index

Let d be an end-of-life device with state vector $s(d)$. Let r be a candidate retention strategy in the R-hierarchy, and let $F(r)$ be the set of licensed and equipped facilities able to perform r . The Execution Feasibility Index $\varphi(d, r)$ is the product of five terms, each bounded on the unit interval:

$$\varphi(d, r) = A(r) \times \text{Acc}(d, r) \times \text{Cap}(r, t) \times V(d, r) \times L(d, r) \tag{1}$$

where the terms are as follows.

- $A(r)$ is the availability term, a binary indicator equal to 1 if $F(r)$ is non-empty within the administratively reachable system and 0 otherwise. The binary form is deliberate. Existence of a licensing category is not the same as availability of a facility able to accept a particular device, so $A(r)$ as specified here is a permissive upper bound.
- $\text{Acc}(d, r)$ is the accessibility term, a monotonically decreasing function of the effective transport distance from the device origin to the nearest facility in $F(r)$, modulated by route quality, consolidation opportunity, and lawful transport constraints on hazardous fractions. We adopt a logistic-decay form, $\text{Acc} = 1 / (1 + \exp[\beta(\delta - \delta_{1/2})])$, where δ is effective road distance, $\delta_{1/2}$ is the distance at which accessibility falls to one half, and β controls the steepness of decay. Both parameters are device-class specific: decay is steep for low-mass, low-unit-value items and shallow for high-value or high-mass items over which transport cost is amortised against recoverable value.
- $\text{Cap}(r, t)$ is the capacity term, the probability that a facility in $F(r)$ can accept the device within the value-retention window beginning at time t . It is computed from installed throughput, current utilisation, and expected queue length. Of the five terms, this one is most amenable to real-time instrumentation, and it is the one most conspicuously absent from current systems.
- $V(d, r)$ is the viability term, a normalised measure of the net economic surplus generated by executing r on d after collection, aggregation, transport, processing, compliance, and margin requirements at every intermediate node. V must be read as the minimum across the chain, not the sum across it: a chain with positive aggregate surplus but one value-negative intermediary will not run, because that intermediary defects to the informal channel. Formally, V is zero if any required actor faces a non-positive surplus.
- $L(d, r)$ is the legibility term, the probability that a completed transaction will be recorded and credited within the prevailing institutional framework. It is zero along routes passing through unregistered actors. In India, legibility failure and viability success are frequently correlated: the least legible path is often the cheapest executable one.

3.4 Aggregate measurement: the Circularity Execution Ratio

The Execution Feasibility Index evaluates a single device–strategy pair. System-level assessment requires an aggregate. Let $R^*(d)$ be the strategy an unconstrained optimiser would recommend for device d on purely technical grounds, that is, the highest attainable point on the R-hierarchy. Let $\hat{R}(d)$ be the highest-ranked strategy whose φ exceeds a decision threshold. For a population of devices D we define the Circularity Execution Ratio as:

$$\text{CER}(D) = \frac{\sum_{d \in D} w(d) \cdot \text{value}(\hat{R}(d))}{\sum_{d \in D} w(d) \cdot \text{value}(R^*(d))} \tag{2}$$

where $w(d)$ weights devices by mass, embodied impact, or recoverable value according to the purpose of the analysis, and $\text{value}(\cdot)$ maps a strategy to a retained-value or avoided-impact score. A system with no execution gap has a CER of one; a system whose recommendations are systematically inexecutable has a CER approaching zero. A summary measure of the AI–Physical Execution Gap follows directly as

$$AIPEG(D) = 1 - CER(D).$$

The measure is useful for policy in three respects. Because φ is multiplicative, shortfalls can be attributed to specific dimensions, so a state or district can see which constraint to relieve first. CER can be computed by district, which maps the geography of inexecutable circularity and identifies where marginal infrastructure investment would release the greatest volume—arguably the highest-value use of AI in an infrastructure-constrained circular economy. The comparison is also counterfactual: realised outcomes against technical potential, rather than against last year’s performance, which is the relevant benchmark for a resource-constrained transition.

3.5 The status of these measures

φ and CER are proposed measurement instruments, not validated field indices. Section 6.3 reports an illustrative computation under Appendix B. Section 6.5 implements those same equations on a 30-origin Indian geographic panel. Neither exercise is a census of Indian circularity: accessibility decay parameters remain uncalibrated against operator logs, and the hub file is a literature-informed proxy for the unpublished geocoded CPCB register. The results test the *structure* of the theory. Section 9.4 sets out what a first empirical estimation from the live portal would require.

4. India’s Material Substrate: What the Algorithm Is Recommending Into

4.1 Volume, composition, and trajectory

International monitors and the Government of India do not report the same Indian e-waste volume, and the gap is large enough to be a finding rather than a footnote. The Global E-waste Monitor 2024 lists India at 4.1 million tonnes in 2022, third in the world after China (Baldé et al., 2024). NITI Aayog, citing the same ITU/UNITAR family, reports a rise from about 2.76 million tonnes in 2020 to about 6.19 million tonnes in 2024 and a projection of 14 million tonnes by 2030 (NITI Aayog, 2026). A market assessment by RedSeer (2025) gives 3.8 million tonnes for FY 2024 and likewise ranks India third. CPCB estimates, built from registered producers’ sales and prescribed average lives under the 2022 Rules, are 1,254,286.55 tonnes in FY 2023–24 and 1,397,955.59 tonnes in FY 2024–25 (MoEFCC, 2025). Formal collection-and-processing shares against that official denominator were 61.94 percent and 70.71 percent in those two years (MoEFCC, 2025).

Composition matters more than the headline tonne count for this chapter. India recorded among the fastest growth rates in screens, computing equipment, and small IT and telecommunication equipment (Baldé et al., 2024). TRAI counted 1,150.66 million wireless subscriptions on 31 December 2024, of which 523.58 million were rural (TRAI, 2025). A smartphone typically weighs 180–200 grams and concentrates gold, silver, palladium, copper, cobalt, lithium, tantalum, and rare-earth elements in the printed circuit board. Urban-mining studies have long noted that such boards can be richer per tonne than many primary ores (Zeng, Mathews, & Li, 2018). That is the device class that most rewards high-R strategies and most penalises unconsolidated reverse logistics.

4.2 Evidence of data asymmetry

The series in Section 4.1 diverge by a factor of about three to five, depending on which international figure is used as the numerator. That is not an error to be averaged away. CPCB measures against an official sales-based denominator; the country loses material against the apparent-consumption denominator used by Baldé et al. (2024). The ratio is a crude proxy for the **legibility** dimension of AIPEG. Formal processing shares that climb from 61.94 to 70.71 percent against 1.25–1.40 million tonnes can be reported as a compliance success while independent estimates continue to place most physical handling outside the portal. ICEA (2024) attributes about 90 percent of collection volume and 70 percent of recycling volume to the informal sector. Turaga et al.

(2019) similarly report that more than 90 percent of Indian e-waste has been processed informally. NITI Aayog (2026) still describes the informal sector as dominant (about 62 percent in its synthesis). Rising portal rates against a smaller official base, alongside a persistently large informal share on independent measures, is phantom circularity: improvement in the measurement apparatus read as improvement in material outcomes. An AI system trained only on portal data inherits that boundary.

4.3 Composition and scarcity of facilities

As of 9 February 2025 the CPCB EPR portal listed 322 registered recyclers with reported capacity of 2,208,918.064 tonnes a year and 72 registered refurbishers with 92,042.18 tonnes a year (Press Information Bureau, 2025). The ratio is 4.47 recyclers per refurbisher. NITI Aayog (2026) notes that 30 high-capacity recyclers controlled over 60 percent of formal processing capacity in the ICEA compilation it reports. Density is thin relative to 1.44 billion people (United Nations DESA, 2024) and more than a billion wireless subscriptions (TRAI, 2025). The register is also biased toward the lower R-strategies: even a perfect optimiser recommending refurbishment at scale would face an execution layer more than four times thinner than the recycling layer. Licensing categories are built around dismantling and material recovery. Component harvesting, repair-grade parts qualification, and module remanufacturing have no dedicated home, so the intermediate strategies that retain the most value for semiconductor devices have no reliable formal executor.

Capacity is spatially concentrated. NITI Aayog (2026) cites 2,808 collection centres for the national population. Generation is not. A large share of arisings originates outside the metropolitan cores that host processing plants, including districts several hundred kilometres from any licensed facility. That mismatch is the material basis of the accessibility dimension. Section 6.5 measures it on a 30-origin panel.

4.4 The shadow execution layer: the informal sector

If official formal shares are 61.94–70.71 percent of the CPCB denominator, residual official-informal volume on that denominator is about 29–38 percent. Independent estimates are higher, as noted above (Turaga et al., 2019; ICEA, 2024; NITI Aayog, 2026). The analytical point is the functional role, not the exact percentage. The informal sector is a second execution layer, stronger on accessibility, viability, and elastic capacity, and near zero on legibility and on environmental and occupational quality. Open burning, acid leaching, and manual desoldering recover part of the available metal while imposing health externalities on workers and nearby communities (Chatterjee & Kumar, 2009; Wath et al., 2011; Awasthi & Li, 2017). ICEA (2024) notes that informal collectors can offer higher prices because their cost structure excludes GST and EHS compliance—the viability channel in our terms.

Across much of India this layer is the only actor that performs last-mile collection. An AI architecture that treats informality only as a compliance defect will recommend pathways that cannot run outside a handful of clusters. An architecture that treats it as an execution asset to be selectively formalised faces a different feasibility frontier. Sections 7 and 8 take up that design choice.

5. Stage-by-Stage Bottleneck Analysis: The Smartphone Circular Chain

5.1 Stage 1: Discard decision and collection

The household decision is where residual-value models are usually asked to work. They price the handset, point the owner toward a formal channel, and quote a buy-back. None of that is execution. Execution still requires a collection point within a distance the owner will actually travel, or a reverse pickup that beats the cash offer at the door. Outside metropolitan India those two conditions are rarely both present. Take-back lives inside organised retail and brand service networks; those networks are urban. Formal quotes lose to immediate cash.

Many retired handsets then sit in drawers for years, and residual value decays before any downstream plant is even in play. Accessibility and viability bind at this stage, well before capacity at a recycler can matter.

5.2 Stage 2: Aggregation and consolidation

A 180-gram device does not pay its own way as a singleton shipment. Reverse logistics works only when loads are bundled. Routing software can draw pickup paths and name consolidation points; it cannot invent a shed, a lock, or a custody trail. Below the district, formal aggregation is thin or missing. Informal *kabadiwalas* and small traders already perform that function, but they are outside the digital system and therefore outside the optimiser's world model. Accessibility failure and legibility failure meet here. For most Indian e-waste, this is the point of irreversible exit from the formal pathway.

5.3 Stage 3: Triage and grading

Machine vision and diagnostic software can assign an initial R-strategy, grade cosmetic condition, and verify component functionality. This is the most substrate-coupled stage, and it is where AI currently adds the most demonstrable operational value. It also requires a facility with imaging apparatus, diagnostic rigs, and trained operators. India's formal triage capacity is largely confined to a small number of refurbishment and trade-in operations. Grading standards are not nationally harmonised, so a grade assigned at one node cannot be read as equivalent to a grade assigned at another. That is a technical form of legibility failure, and a direct obstacle to distributed triage.

5.4 Stage 4: Reconditioning and refurbishment

For a functional handset with a degraded battery, refurbishment is almost always the value-maximising strategy, because it retains the device's full embodied impact. Lawful, warrantied resale requires genuine or qualified spare parts, service documentation, diagnostic access, and an authorised entity. Availability binds immediately and hard: there were 72 registered refurbishers nationally as of 9 February 2025 (Press Information Bureau, 2025). Manufacturers' parts policies further restrict the supply of spares for older models, and repairability varies considerably across device designs. India's evolving right-to-repair framework bears directly on this stage, because it acts on the constraint that determines whether the highest-value R-strategy is executable at all.

5.5 Stage 5: Component harvesting

Where full refurbishment is not feasible, salvaging display assemblies, camera modules, connectors, and memory for the repair market yields substantially more value than shredding. Component-level value estimation is a pattern-recognition problem at which models excel, and it is among the strategies AI systems most readily recommend. It is also the strategy with the thinnest formal implementation in India. Harvesting occupies a regulatory interstice between dismantling and refurbishment, has no dedicated authorisation category, and has no qualification standard that would allow harvested parts to enter formal repair channels under warranty. Informal clusters, meanwhile, harvest components at scale and with considerable skill. The capacity exists, but it is illegible and unqualified, and is therefore unavailable to the formal system.

5.6 Stage 6: Dismantling and fraction separation

Manual and semi-automated dismantling separates battery, display, board, housing, and cable fractions for divergent downstream processing. This is where India's formal capacity is genuinely concentrated and where authorised dismantlers operate. Binding constraints here are capacity and throughput rather than availability, together with a technical constraint rarely encoded in recommendation systems: lithium-ion batteries are classified as hazardous for transport, and devices containing them are subject to routing restrictions that a naïve distance-minimising optimiser will violate.

5.7 Stage 7: Refining and material recovery

Recovering gold, silver, palladium, copper, and cobalt from board fractions requires hydrometallurgical or pyrometallurgical capability, a technical standard only a fraction of registered recyclers meet. India’s integrated capacity to refine high-value precious metals from electronic scrap is limited, and part of the highest-value fraction is exported for refining—an outcome that satisfies material circularity in the global aggregate while forgoing domestic value capture. Rare-earth elements are recovered from e-waste at a negligible rate globally: the Global E-waste Monitor reports that about 1 percent of rare-earth demand is met by e-waste recycling (Baldé et al., 2024). At this stage the binding constraints are capacity and viability, and the limiting factor is the sophistication of installed technology rather than the number of registered facilities.

5.8 Stage 8: Resale of recovered materials and refurbished devices

The loop closes only when recovered materials and refurbished devices re-enter production and consumption. India has a large and growing secondary smartphone market, which is a genuine structural advantage: demand for value-retained devices is not the limiting factor. Trust and standardisation are. A refurbished handset without a credible grade, warranty, and provenance record attracts a steep discount, and that discount cascades backward through the chain, destroying the viability of the refurbishment pathway that produced the device. A digital product passport architecture would generate a disproportionate return at this stage precisely because it addresses a legibility failure at the point where value is realised rather than where material is handled.

5.9 Synthesis

Table 2. Stage-wise comparison of algorithmic capability and formal execution capability in India’s smartphone circular chain.

Stage	AI capability maturity	Formal execution capability	Binding gap dimension
1. Discard and collection	High	Low outside metropolitan areas	Accessibility, Viability
2. Aggregation	High	Very low; informally dominated	Accessibility, Legibility
3. Triage and grading	Very high	Low; unstandardised	Availability, Legibility
4. Repair and refurbishment	Moderate	Very low (72 registered entities)	Availability
5. Component harvesting	High	Near zero formally; high informally	Availability, Legibility
6. Dismantling	Moderate	Moderate; spatially concentrated	Capacity, Accessibility
7. Material recovery	Moderate	Limited at high-value tiers	Capacity, Viability
8. Resale	High	Moderate and improving	Legibility

Read across the table and a consistent pattern appears: algorithmic capability is highest exactly where formal execution capability is lowest. Triage, harvesting, and residual-value estimation—the tasks models perform best—depend on refurbishment and harvesting layers that barely exist in India’s formal system. Conversely, the stages where the marginal AI contribution is smallest, dismantling and bulk material recovery, are the stages where India holds real capacity. The execution gap is not spread evenly along the chain; it is concentrated where the algorithmic promise is greatest. Intervention strategies that overlook this inverse relationship will misdirect both computational and capital investment.

6. The Urban–Rural Execution Divide

6.1 Distance, mass, and consolidation

Whether circularity is worth performing on a low-mass, high-value device is governed by the cost of moving that device to a capable facility. That cost has two components with very different behaviour. One is a haulage

component, roughly proportional to distance and to mass. The other is a fixed component per consignment (vehicle call-out, stop time, handling, custody, and documentation) that is incurred once per shipment regardless of how many devices the shipment carries, and is therefore divided across the consolidation factor. Recoverable value per device, meanwhile, is bounded and modest for a mid-range end-of-life smartphone. There is consequently a break-even distance beyond which moving a device costs more than the value recoverable from it and, independently, a distance beyond which transport emissions exceed the emissions avoided by the circular intervention. Above either threshold the recommendation is not merely uneconomic but counterproductive.

Rural circularity is not simply a harder version of urban circularity. The same device generates a different cost structure depending on whether a consignment can be assembled around it. An optimiser that represents neither distance nor consolidation will systematically recommend value-destroying actions across most of India's geography while reporting them as circular successes.

6.2 A break-even analysis

Let the net circular surplus of routing one device from an origin to a capable facility be:

$$S = V_{\{rec\}} - C_{\{acq\}} - C_{\{agg\}} - C_{\{proc\}} - C_{\{comp\}} - C_{\{cons\}}/n - c_{\{t\}} \cdot \delta \cdot m \tag{3}$$

where $V_{\{rec\}}$ is recoverable value, $C_{\{acq\}}$ the acquisition price paid to the holder, $C_{\{agg\}}$ aggregation and handling cost, $C_{\{proc\}}$ processing cost, $C_{\{comp\}}$ compliance and documentation cost, $C_{\{cons\}}$ the fixed cost of assembling and dispatching one consignment, n the consolidation factor (devices per consignment), $c_{\{t\}}$ the haulage cost per tonne-kilometre, δ the effective road distance, and m the device mass. Writing $M = V_{\{rec\}} - C_{\{acq\}} - C_{\{agg\}} - C_{\{proc\}} - C_{\{comp\}}$ for the margin before transport, the break-even distance is:

$$\delta_{\{be\}} = (M - C_{\{cons\}}/n) / (c_{\{t\}} \cdot m) \tag{4}$$

and the minimum viable consignment size at a given distance is:

$$n_{\{min\}} = C_{\{cons\}} / (M - c_{\{t\}} \cdot \delta \cdot m) \tag{5}$$

Appendix A derives (4) and (5) from (3). Note that (3) departs from the formulation most commonly used in reverse-logistics costing, which expresses transport cost purely as a tonne-kilometre term. For a 180-gram device that formulation is dimensionally correct but empirically misleading, because the mass–distance product is so small that the haulage term becomes negligible against the per-consignment fixed cost. Separating the two components is what allows the model to identify the operative constraint rather than the nominal one.

Four structural properties follow for any reasonable parameterisation.

First, the break-even distance increases with the consolidation factor but is bounded above by $M/(c_{\{t\}} \cdot m)$. For a smartphone that bound is very large. Distance therefore does not bind through haulage cost; consolidation binds, and distance matters chiefly because it determines whether a consignment can be assembled at all within the value-retention window.

Second, at constant value the break-even distance is inversely proportional to device mass, which explains why smartphones are simultaneously the most attractive devices to recover and among the most fragile to route: their high value density supports long-distance movement when consolidated and collapses to a near-zero viable radius when they are handled individually.

Third, consolidation cannot rescue a pathway whose margin before transport is non-positive. Where $M \leq 0$, expression (5) admits no finite $n_{\{min\}}$. Aggregation repairs viability failures caused by cost allocation; it does not repair viability failures intrinsic to the pathway's economics.

Fourth, C_{comp} enters with the same sign as transport cost. Every additional layer of documentation, verification, and compliance imposed on a formal pathway contracts the region in which that pathway is viable and expands the region in which the informal pathway is the only economically rational choice. Compliance design is, unintentionally, spatial policy.

6.3 Three territorial regimes: an illustrative computation

To show what these properties imply, we evaluate (3) and (1) for three territorial regimes and three retention strategies under the parameter assumptions set out in Appendix B. The parameters are plausible order-of-magnitude values drawn from published freight tariffs and secondary-market prices; they are not measured operator data, and the results below are a demonstration of structure, not an estimate of Indian circularity.

Table 3. Illustrative computation of per-device net surplus and Execution Feasibility Index by territorial regime and retention strategy, under the parameter assumptions of Appendix B. Values in Indian rupees per device. These are illustrative computations, not empirical estimates.

Territorial regime	Δ (km)	n	Transport ₹/device	Strategy	M (₹)	S (₹)	Binding term	φ
Metropolitan core	25	200	2.03	Refurbishment	850	848	—	0.70
Metropolitan core	25	200	2.03	Component harvesting	200	198	Availability	0.00
Metropolitan core	25	200	2.03	Material recycling	-35	-37	Viability	0.00
Mid-periphery	180	40	22.71	Refurbishment	850	827	—	0.48
Mid-periphery	180	40	22.71	Component harvesting	200	177	Availability	0.00
Mid-periphery	180	40	22.71	Material recycling	-35	-58	Viability	0.00
Hinterland, no node	520	1	250.59	Refurbishment	850	599	Accessibility	0.01
Hinterland, no node	520	1	250.59	Component harvesting	200	-51	Availability	0.00
Hinterland, no node	520	1	250.59	Material recycling	-35	-286	Viability	0.00
Hinterland with node	520	60	23.93	Refurbishment	850	826	—	0.51
Hinterland with node	520	60	23.93	Component harvesting	200	176	Availability	0.00
Hinterland with node	520	60	23.93	Material recycling	-35	-59	Viability	0.00

Three findings follow. First, across the full range examined, the haulage term contributes between ₹0.03 and ₹0.59 per device, a spread of about ₹0.56 across 495 kilometres, while the consolidation term contributes between ₹2.00 and ₹250.00, a spread of ₹248. The consolidation factor dominates haulage distance by more than two orders of magnitude. The widely repeated claim that distance is the primary obstacle to rural circularity for small devices is, on this parameterisation, incorrect in its mechanism even though it is correct in its conclusion: what makes remote devices unroutable is that no consignment can be assembled around them, not that they are far away.

Second, inserting an aggregation node in the hinterland raises the per-device net surplus of the refurbishment pathway from ₹599 to ₹826 and, more importantly, raises its Execution Feasibility Index from approximately 0.01 to approximately 0.51, because consolidation brings a distant but existing refurbisher into the effectively reachable system. This is the quantitative case for the aggregation node as the highest-leverage infrastructure asset in rural circularity.

Third, and against the grain of that conclusion, consolidation does nothing for component harvesting. Its net surplus is comfortably positive in three of the four regimes, yet its Execution Feasibility Index is zero everywhere, because the availability term is zero: no authorisation category exists for component harvesting anywhere in India. Similarly, formal material recycling of an individual handset is value-negative in every regime, and no consolidation factor repairs it; registered recyclers sustain themselves on mixed bulk streams and certificate revenue rather than on per-handset margin. Aggregation infrastructure and licensing reform are

therefore complements, not substitutes. Building nodes without opening a harvesting category converts an accessibility failure into an availability failure and leaves φ at zero.

This is the answer to RQ2, and it is conditional. Under present institutional arrangements an AI recommendation issued for a rural end-of-life device cannot be executed at a cost and environmental impact that make circularity worthwhile, unless an aggregation layer is interposed that converts individual devices into consolidated flows before the transport cost is allocated, and unless a licensing category exists for the strategy the optimiser selects. Either condition alone leaves the recommendation inexecutable.

6.4 Design implications

This inverts the conventional intervention logic. The standard prescription for rural circularity is to push formal collection and processing capacity outward from the core. The break-even structure shows that to be the more expensive of two options: because per-device transport cost falls as C_{cons}/n , low-capital aggregation nodes—which require storage, a custody protocol, a digital identity, and a payment rail but no processing plant—extend the viable radius far more cheaply than duplicating processing capacity. Such nodes can be hosted within existing rural physical networks of national coverage, including the postal network, cooperative structures, common service centres, and organised retail.

It also redefines what AI should be doing in rural India. Per-device pathway recommendation is not a useful computation there, because it is precisely the computation whose output cannot be acted upon. The useful computation is aggregation-node siting and consolidation scheduling: where to place a node, at what cost, to convert the largest volume of currently inexecutable recommendations into executable volume, and when to dispatch once accumulated volume makes dispatch worthwhile. That is a facility-location and inventory-routing problem under uncertainty, a class of problems with mature methods, and one tractable with data India already collects.

6.5 A reproducible geographic test of the theory

The four-regime arithmetic of Table 3 can be accused of having been chosen to illustrate the argument. We therefore implement equations (1)–(5) in open code and apply them to thirty named Indian origins whose coordinates are public: eight metropolitan cores, ten state capitals treated as mid-periphery collection points, and twelve hinterland districts (including Nuapada, Odisha, the opening vignette). Each origin is assigned a road distance to the nearest of nine literature-informed processing hubs (Delhi-NCR, Moradabad, Bengaluru, Hyderabad, Chennai, Mumbai, Pune, Ahmedabad, Kolkata) using great-circle distance scaled by a road circuitry factor of 1.30. Cost parameters remain those of Appendix B. Accessibility is evaluated on distance to the first formal custody point, so a hinterland aggregation node shortens first-mile distance even when the licensed plant remains far away. Component harvesting remains unlicensed ($A = 0$) in every location. Full tables are in `results/`; figures are generated by `analysis/run_experiment.py`.

Three structural tests of the theory survive this geographic extension.

T1. Consolidation, not haulage, dominates small-device transport cost. On the Appendix B regimes, haulage ranges from ₹0.03 to ₹0.59 per device (spread ₹0.56). The consolidation term ranges from ₹2.00 to ₹250.00 (spread ₹248). The ratio of those spreads is about 443:1, more than two orders of magnitude, reproducing the claim of Section 6.3 on an independent implementation of equation (3).

T2. Availability is non-compensatory. Maximum φ for component harvesting is 0.00 in every regime and every origin, including metropolitan cores where surplus is positive (about ₹198). Formal material recycling of a single handset has negative surplus in every regime (metropolitan $S = -37$), so $V = 0$ and $\varphi = 0$. Nodes do not substitute for a licence, and they do not repair a negative margin. That is the conjunctive structure of equation (1).

T3. Hinterland executability is a node problem. Mean refurbishment φ across the twelve hinterland origins is 0.011 without a node and 0.536 with a node. Using a decision threshold $\varphi > 0.10$, the Circularity Execution Ratio (equation 2) is 1.00 in metropolitan origins and in state capitals with a collection point, 0.00 in hinterland origins without a node, and 1.00 once a node is added—but **only for licensed, value-positive strategies**. Harvesting never enters the executable set. Nuapada, the vignette district, lies about 742 road-kilometres from the nearest hub (Hyderabad) in this proxy network; without a node $\varphi \approx 0$ for every strategy.

These results do not estimate India’s true CER. They show that, given published facility scarcity, public geography, and the chapter’s own cost assumptions, the theory’s qualitative predictions are internally consistent and geographically non-trivial. A live CPCB geocoded register would replace the nine-hub proxy; it would not replace the multiplicative test.

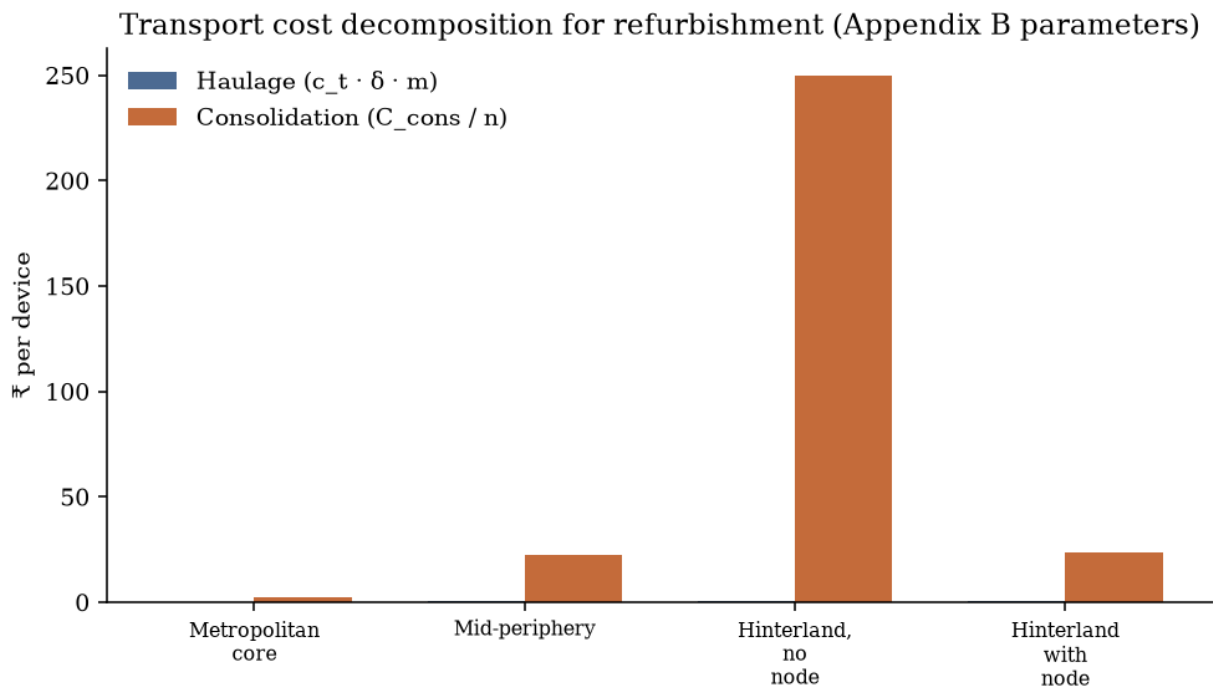


Figure 1. Haulage versus consolidation cost per device for refurbishment under Appendix B.

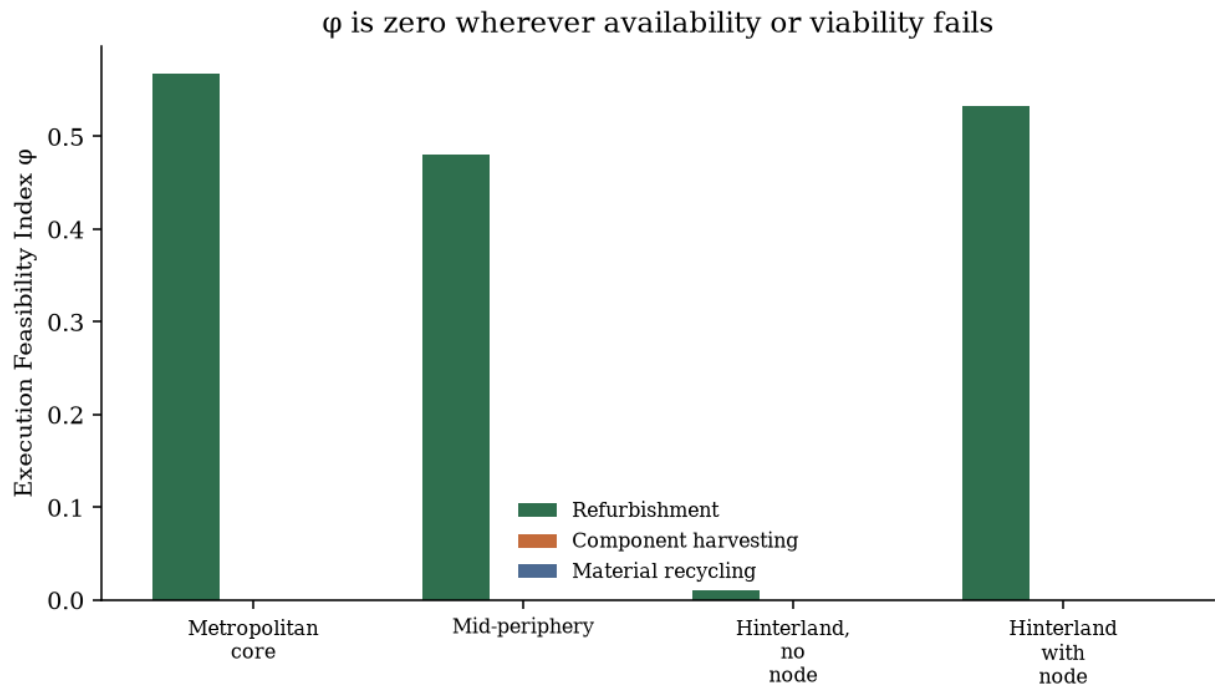


Figure 2. Execution Feasibility Index by territorial regime and strategy. Harvesting and per-handset recycling remain at zero.

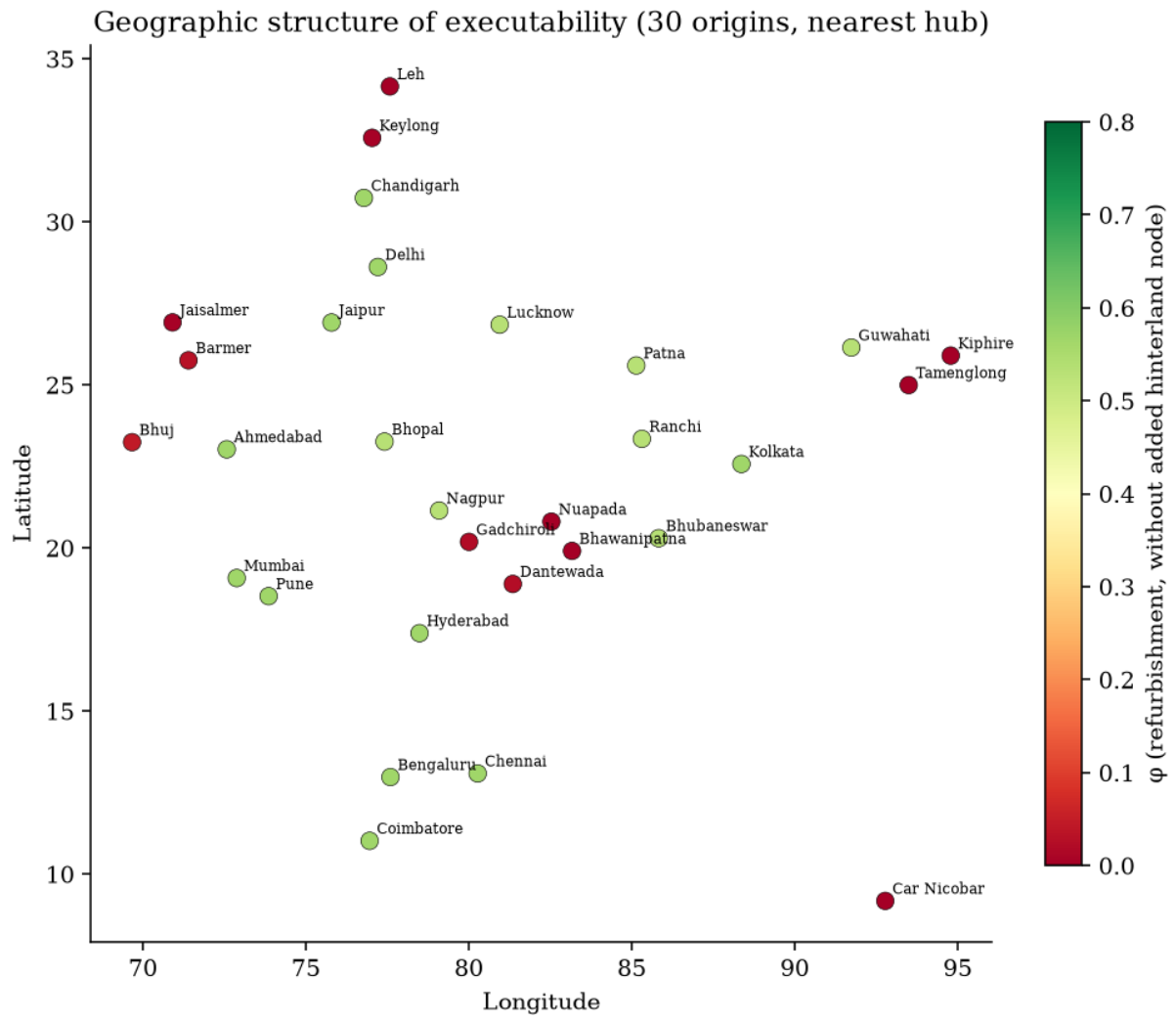


Figure 3. Refurbishment ϕ at thirty origins, without added hinterland nodes.

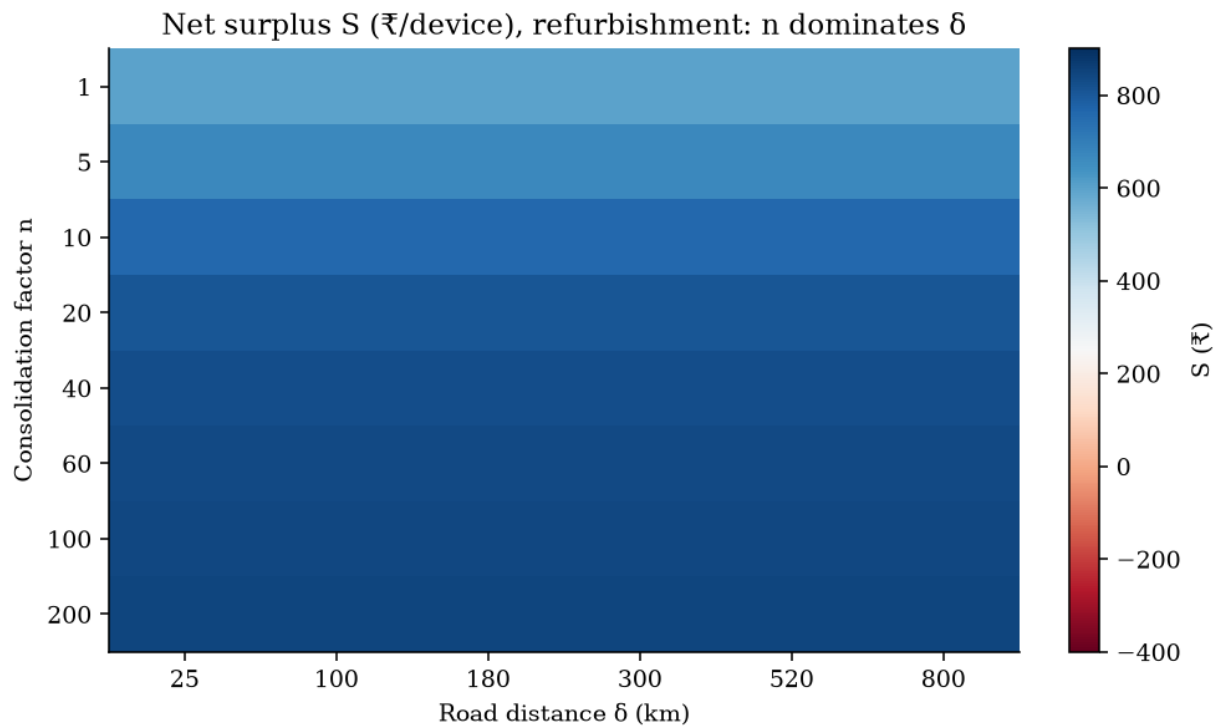


Figure 4. Net surplus S for refurbishment across a grid of n and δ . Variation in n dominates variation in δ .

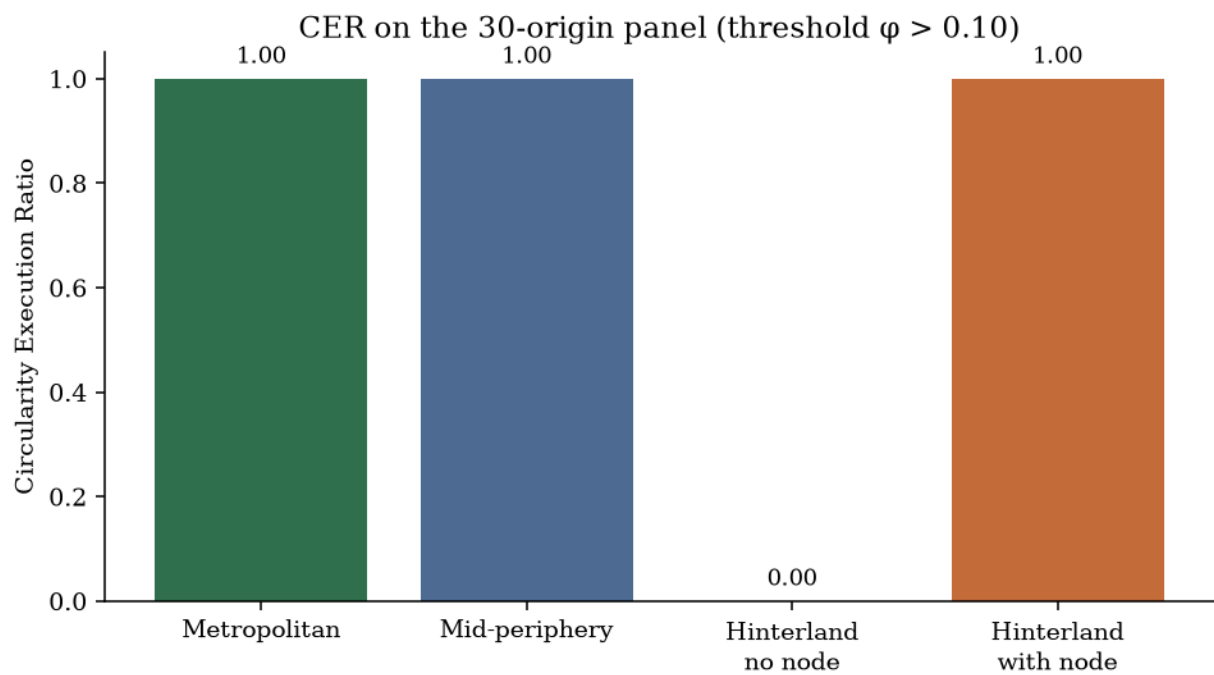


Figure 5. Circularity Execution Ratio on the thirty-origin panel (threshold $\varphi > 0.10$).

7. Institutional, Policy and Market Conditions for Closing the Gap

7.1 What the E-Waste (Management) Rules, 2022 cover, and what they do not

The regime notified in November 2022 and in force from April 2023 is ambitious on paper. Product coverage is wider than under earlier rules. Registration and filing run through a central CPCB portal. Producer recycling targets under the Rules rise from 60 percent in FY 2023–24 and 2024–25 to 70 percent in FY 2025–26 and 2026–27 and 80 percent from FY 2027–28 (Government of India, 2022; Press Information Bureau, 2022). Recyclers earn tradable EPR certificates against quantities recycled; producers buy those certificates to discharge their obligations. Circularity, in this design, is administered as a digital compliance market.

The five executability conditions expose how incomplete that market is. Legibility for registered actors is the dimension the Rules take most seriously: a national data spine now exists where none did. Viability is touched only indirectly, through certificate revenue that improves recycler economics. Availability is untouched. Nothing in a certificate market requires a plant in an underserved district or an authorisation category that does not already exist. Accessibility is not merely neglected; certificates are spatially fungible, so a certificate generated in a high-capacity industrial state can discharge an obligation arising from devices sold in a state with no facilities. Capacity signalling is absent as well. The portal records completed transactions, not unused throughput.

Spatial fungibility has a predictable equilibrium. The cheapest discharge of an obligation is a certificate bought from the lowest-cost recycler, and that recycler is already where capacity is concentrated. The market therefore pays for clustering. Accessibility worsens as a designed outcome, not as an oversight. Measurement in processed tonnes, rather than retained value, produces a second equilibrium: refurbishment and shredding are compliance-equivalent. A mass-recycling metric cannot preferentially pull activity up the R-hierarchy. The four-and-a-half-to-one ratio of recyclers to refurbishers in Section 4.3 is consistent with that incentive, not an accident of industrial history.

7.2 Digital public infrastructure as a coordination substrate

India's digital public infrastructure is the most credible existing asset for addressing the legibility and viability dimensions together: a population-scale identity layer, a real-time low-cost payment rail, a consent-based data-sharing architecture, and an open network protocol for digital commerce. Its relevance to circularity remains largely unexplored.

The mechanism is specific. The exclusion of informal aggregators from the EPR system is not a matter of intent but of transaction infrastructure: registration is costly, invoicing presumes formal accounting, and a handover cannot be paid for and evidenced at low cost. A tiered digital identity for aggregators, instant account-to-account settlement on handover, and a lightweight cryptographic handover receipt would convert an illegible transaction into a legible one without requiring the aggregator to re-engineer their business. An open network protocol layer could then publish facility capacity, current buy-back prices, and accepted device categories as discoverable machine-readable offers: exactly the data the capacity and viability terms of the Execution Feasibility Index require, and exactly the data no current system publishes.

The caveat is the one this chapter exists to press. Digital public infrastructure improves coordination, and coordination raises the productivity of existing capacity. It does not create capacity. A perfectly instrumented market containing 72 registered refurbishers will allocate those 72 refurbishers efficiently and will still be unable to refurbish at national scale. Digital coordination is necessary; it is plainly not sufficient.

7.3 Five institutional conditions

In summary, the execution gap can be compressed only by the joint application of five institutional conditions.

- **Spatially differentiated compliance.** Certificate value should be modulated by the origin of the device rather than the location of processing, so that discharging an obligation with material collected from an underserved district is worth more than discharging it with metropolitan material. This converts a spatially blind market into an instrument for funding access.
 - **Strategy-differentiated compliance.** Certificate value should be modulated by the level of value retention, so that material recycling is rated below refurbishment and qualified component harvesting. Without this, no market signal will correct the availability distortion at the top of the R-hierarchy.
 - **A licensing category for component harvesting and parts qualification.** The intermediate strategies that retain the most value for semiconductor-dense devices currently have no regulatory home, and the informal capacity that already performs them cannot be brought into the formal system without a category to bring it into.
 - **Formalisation of the aggregation layer.** A registration tier adapted to the real operating conditions of small aggregators (low documentation burden, digital identity, verified handover, guaranteed settlement) would capture the accessibility advantage of the informal network rather than competing with it. Indian experience with the integration of waste pickers into municipal solid waste systems provides precedent, and the evidence there indicates that inclusion outperforms displacement (Chintan, 2014; Wilson et al., 2006).
 - **A public capacity register.** Facility throughput, current utilisation, accepted categories, and buy-back prices should be published in machine-readable form in the public domain. This is the precondition for the capacity term of any execution-aware optimiser, and it is the cheapest of the five conditions to implement.
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8. Designing Circular AI for Infrastructure-Constrained Settings

8.1 A shift in framing

Most circular-AI systems still rank R-strategies first and treat infrastructure as something to be “implemented” after the ranking is done. **Infrastructure-Constrained Circular AI (ICCAI)** inverts that sequence. The state of the execution environment enters the objective function before ranking, and search is restricted to the executable feasible set rather than the technically ideal set. The output is not the best thing that could be done to the device in an unconstrained world. It is the best thing that can be done to this device, from this place, given the plants that exist, the throughput they have today, and the surplus the chain can actually share.

8.2 Architecture

ICCAI comprises four layers.

The **inference layer** performs conventional device-state estimation: classification, condition rating, component functionality assessment, residual-value prediction, and technical R-strategy ranking. This layer is analogous to existing practice, and existing models can be reused within it directly.

The **constraint layer** is the novel component. It maintains a live model of the execution environment and computes the Execution Feasibility Index for each candidate device–strategy pair. Table 4 sets out the variables it must carry, their plausible sources, and the gap dimensions each addresses.

Table 4. Constraint-layer variables, data sources, and the gap dimensions they address.

Constraint variable	Data source	Update frequency	Gap dimension addressed
Facility register with licensed strategies	CPCB EPR portal; state pollution control boards	Weekly	Availability
Facility geolocation and road-network distance	Public geospatial data; routing services	Static, with route updates	Accessibility
Real-time utilisation and queue depth	Facility-reported capacity register	Daily to hourly	Capacity
Transport tariffs, consolidation state, hazardous-routing rules	Logistics providers; transport regulation	Daily	Accessibility, Viability
Secondary market prices and buy-back quotes	Marketplaces; open network protocol catalogues	Hourly	Viability
Aggregator network state and coverage	Tiered registration and handover records	Daily	Accessibility, Legibility
Compliance and documentation cost by pathway	Regulatory schedules; portal fee structures	Quarterly	Viability, Legibility

The **decision layer** ranks strategies by expected executed value, that is, technical retained value multiplied by the Execution Feasibility Index, rather than by technical value alone. It emits a pathway, a designated facility, an aggregation route, a dispatch window, and an explicit feasibility score. Where no strategy clears the feasibility threshold, it returns a deferral with a stated reason: hold pending consolidation, hold pending capacity, or route to the highest-integrity channel available with the binding constraint recorded. A deferral is not a recommendation, and the distinction matters for reporting.

The **feedback layer** closes the loop that present systems leave open. Every recommendation is linked to a verified outcome, and every deviation is recorded as an execution failure attributed to a specific gap dimension. This telemetry is the raw material of the aggregate Circularity Execution Ratio, and it is what converts a recommendation system into a diagnostic instrument for infrastructure planning.

8.3 Five design principles

Principle 1: Constrain, then rank. Compute feasibility first and rank desirability within the feasible set. A rank-first, filter-second architecture will systematically over-promise, because the filter stage is invariably weaker than the ranking stage and is frequently absent from deployment altogether.

Principle 2: Geographical realism. Distance, route quality, hazardous-materials routing restrictions, and consolidation state must enter the objective function directly. A system that assumes a spatially uniform facility network will generate recommendations that are correct in the metropolitan core and systematically wrong outside it. That, we argue, describes most circular AI currently deployed in India.

Principle 3: Temporal realism about value and capacity. A device whose residual value halves in three weeks cannot be assigned to a pathway that becomes available in six. Capacity should be modelled as a stochastic resource indexed by time, not as a static property of a facility.

Principle 4: Mixed-channel routing. The optimiser's action space must admit informal and semi-formal actors as first-class nodes with their own feasibility profiles, rather than excluding them by definitional fiat. A model restricted to routes between registered entities is blind to the actors who perform most of the collection in the country, and its recommendations are therefore void by construction across most of the country's geography.

Principle 5: Reflexive execution telemetry. The system should measure and expose its own execution gap. This is the principle of greatest institutional value and least technical difficulty: a national Circularity Execution Ratio disaggregated by district, device class, and gap dimension would convert a diffuse

infrastructure debate into a targeted investment problem, and would make phantom circularity visible in the one place it currently is not: the reporting layer.

8.4 Distributional and governance considerations

This design raises three governance problems. Feasibility-aware optimisation is conservatively biased against underserved regions: the hinterland has low φ , so an expected-value optimiser will rationally deprioritise it, potentially entrenching the very inequity the framework diagnoses. This must be countered explicitly, either by weighting the objective for equity or by separating the operational optimiser from a planning optimiser whose objective is to maximise the gain in feasibility per rupee of infrastructure investment. Second, routing to informal actors raises legitimate concerns about environmental and labour standards; the design response is graduated integration with verified handover and progressive substitution of processing functions, not unconditional inclusion. Third, capacity registers and handover records contain commercially sensitive and personally identifiable data, and the consent architecture governing them should conform to the data-minimisation and purpose-limitation norms already established in Indian digital public infrastructure practice.

9. Implications, Limitations and Future Research

9.1 Theoretical implications

This chapter adds an explicit construct of executability to circular-economy theory, and in doing so argues that the R-hierarchy is an incomplete decision framework, because it orders strategies without reference to the systems that execute them. It also offers a counterpoint to the AI-for-sustainability literature: model accuracy and system outcome are not monotonically related, and the relationship can flatten entirely in infrastructure-constrained settings. The AI–Physical Execution Gap generalises beyond e-waste to any domain in which algorithmic capacity has outstripped physical capacity: precision agriculture without cold chains, diagnostic AI without treatment capacity, credit scoring without enforcement infrastructure. We propose, as a hypothesis for testing, that capable inference paired with incapable execution is the characteristic form of the AI transition in emerging economies.

9.2 Implications for practice and policy

For producers and platform operators, circular targets defined over recommendation volume or certificate acquisition are not evidence of circular performance; the quantity that should be governed is a Circularity Execution Ratio computed against verified outcomes. For technology developers, the highest-value application of AI in India’s circular economy is not per-device pathway recommendation but infrastructure planning: siting aggregation nodes, scheduling consolidation, and identifying where marginal capacity converts the greatest volume of currently inexecutable recommendations into executable ones. For regulators, the two highest-leverage and lowest-cost measures identified here are spatial and strategy differentiation of EPR certificate value and a public machine-readable capacity register. Both address structural distortions that no degree of algorithmic sophistication can compensate for.

9.3 Limitations

Three limitations qualify these claims. First, φ and CER remain uncalibrated against operator logs; Section 6.5 tests structural predictions on public geography and Appendix B prices, not a census of Indian circularity. Second, the nine processing hubs are proxies for an unpublished geocoded CPCB register; substituting the live register would change distances, not the multiplicative logic. Third, the analysis is specific to smartphones. Large appliances, with different mass-to-value ratios, will exhibit different break-even structures and possibly a different rank-ordering of binding constraints.

9.4 Future research

Five lines of work follow. First, empirical estimation of the Execution Feasibility Index using facility registry data, road-network distances, and operator cost data, to produce the first district-level map of executable circularity in India. Second, calibration of the accessibility decay parameters by device class against observed reverse-logistics behaviour. Third, controlled comparison of conventional and constraint-aware recommendation systems on the same device population, measuring divergence in realised rather than recommended outcomes. Fourth, treatment of aggregation-node interventions as natural experiments, in order to test the consolidation predictions of Section 6.2 directly. Fifth, comparative application of the framework to other emerging markets (Nigeria, Indonesia, Brazil) to establish whether the structure of the gap is India-specific or a general property of infrastructure-constrained circular transitions.

9.5 Consequences for circular business models

The execution gap reshapes which circular business models can operate viably in an emerging market. Three consequences follow for firms. Product-as-a-service, leasing, and take-back models depend on reverse logistics whose useful radius is set by the consolidation factor rather than by customer demand, so a subscription model that is profitable in a metropolitan cluster may be structurally unprofitable two hundred kilometres away in the same country. Because certificate-based compliance is spatially fungible and indifferent between refurbishment and shredding, producers face no market incentive to develop the higher-value models the R-hierarchy prefers: a firm that invests in refurbishment capacity earns no compliance premium over one that simply buys recycling certificates. And the highest-return position for a circular venture in India may not be processing at all but aggregation: a low-capital, digitally legible intermediary that converts inexecutable recommendations into executable volume and monetises the spread between informal cash offers and formal buy-back prices.

9.6 Relevance to the Sustainable Development Goals

The framework bears most directly on SDG 12, in that it separates reported circular intent from verified circular execution and supplies a measure, the Circularity Execution Ratio, that resists phantom circularity in national reporting. It bears on SDG 9 in that its principal concrete output is a district-level map identifying where marginal infrastructure investment would convert the greatest volume of currently inexecutable circularity into executable circularity. And it bears on SDG 8 in that it treats the informal sector as an execution asset to be graduated into the formal system rather than displaced from it.

10. Conclusion

The opening vignette was a working smartphone in a village in Odisha and a model that could name, in under a second, what ought to happen next. Inference of that kind is now cheap in India. Physical execution is not. Much of the AI-and-circularity literature has treated the first as if it were the second.

We do not argue for delaying AI until every district has a plant, nor for building plants first and models later. The useful change is in the objective. Infrastructure-Constrained Circular AI puts facility availability, distance, transport economics, and live capacity inside the optimiser, and it records unused recommendations as execution failures rather than as circular achievements. Fewer pathways will be recommended. More of those that are recommended will be ones a territory can actually run. Where none can, the honest output is a dated deferral together with the binding constraint.

The last mile will settle India's circular transition: the aggregation shed, the consolidated load, the qualified spare, and the aggregator who can be paid and counted. Algorithms are well suited to locating, district by district, where that last mile is missing. That is not, at present, the question they are being asked.

Declarations

Funding. The authors received no specific funding for this work.

Conflicts of interest. The authors declare no competing interests.

Data availability. Published indicators used in Section 4 are tabulated with sources in `data/published_indicators.csv`. Origin and hub coordinates for Section 6.5 are in `data/district_origins.csv` and `data/facility_hubs.csv`. The computational implementation of equations (1)–(5) is `analysis/aipeg_model.py`; the experiment that writes Tables 3-equivalent results, the district panel, and Figures 1–5 is `analysis/run_experiment.py`. No confidential operator data were used. Appendix B parameters remain stated assumptions.

Declaration on the use of generative AI. The authors used generative AI tools to assist with language editing and structural revision of the manuscript. All conceptual content, formal derivations, parameter choices, interpretations, and conclusions are the authors' own, and the authors take full responsibility for the content of the published work.

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Appendix A. Derivation of the Break-Even Distance and the Minimum Viable Consignment

Equation (3) gives the net circular surplus of routing one device from an origin to a capable facility as

$$S = V_{\text{rec}} - C_{\text{acq}} - C_{\text{agg}} - C_{\text{proc}} - C_{\text{comp}} - C_{\text{cons}}/n - c_{\text{t}} \cdot \delta \cdot m$$

Collect the terms that are invariant with respect to distance and consolidation into a single margin term,

$$M \equiv V_{\text{rec}} - C_{\text{acq}} - C_{\text{agg}} - C_{\text{proc}} - C_{\text{comp}}$$

so that

$$S = M - C_{\text{cons}}/n - c_{\text{t}} \cdot \delta \cdot m \tag{A.1}$$

The break-even distance δ_{be} is the distance at which $S = 0$ for a given consolidation factor. Setting (A.1) to zero and solving for δ :

$$0 = M - C_{\text{cons}}/n - c_{\text{t}} \cdot \delta_{\text{be}} \cdot m \implies c_{\text{t}} \cdot \delta_{\text{be}} \cdot m = M - C_{\text{cons}}/n$$

$$\delta_{\text{be}} = (M - C_{\text{cons}}/n) / (c_{\text{t}} \cdot m) \tag{A.2}$$

reproducing equation (4).

Expression (A.2) is defined and positive only where $M > C_{\text{cons}}/n$. Taking the limit as consolidation increases without bound gives the upper envelope of the viable radius,

$$\lim_{n \rightarrow \infty} \delta_{\text{be}} = M / (c_{\text{t}} \cdot m) \tag{A.3}$$

which establishes the first structural property of Section 6.2: the break-even distance is increasing in n but bounded above by a quantity that depends only on the margin, the haulage tariff, and the device mass. Because m is small for a smartphone, this bound is large, and the haulage term is therefore not the operative constraint over any plausible domestic distance.

The minimum viable consignment size at a fixed distance follows by setting (A.1) to zero and solving for n instead:

$$C_{\text{cons}}/n_{\text{min}} = M - c_{\text{t}} \cdot \delta \cdot m \implies n_{\text{min}} = C_{\text{cons}} / (M - c_{\text{t}} \cdot \delta \cdot m) \tag{A.4}$$

reproducing equation (5).

Expression (A.4) admits a finite positive solution only where $M > c_{\text{t}} \cdot \delta \cdot m$. Where $M \leq 0$, no consolidation factor renders the pathway viable, which establishes the third structural property:

aggregation repairs viability failures arising from the allocation of fixed cost, but not viability failures intrinsic to the margin of the pathway itself.

Finally, differentiating (A.2) with respect to C_{comp} gives $\partial \Delta_{\text{be}} / \partial C_{\text{comp}} = -1/(c_{\text{t}} \cdot m) < 0$, and differentiating with respect to the haulage tariff gives $\partial \Delta_{\text{be}} / \partial c_{\text{t}} < 0$ likewise. Compliance cost and transport cost therefore enter the viable radius with the same sign, which establishes the fourth structural property.

Appendix B. Parameter Values Used in the Illustrative Computation of Section 6.3

The values below are plausible order-of-magnitude figures assembled from published road freight tariffs, observed Indian secondary-market handset prices, and reported informal buy-back offers. They are stated assumptions adopted to demonstrate the structure of the framework. They are not measured operator data, and no empirical claim is made for them. All monetary values are in Indian rupees per device unless otherwise noted.

Table B1. Strategy parameters.

Strategy	V_{rec}	C_{acq}	C_{agg}	C_{proc}	C_{comp}	$M = \text{margin before transport}$
Refurbishment	3000	1200	150	700	100	850
Component harvesting	1400	700	150	250	100	200
Material recycling	120	60	30	40	25	-35

Table B2. Territorial regime parameters. Common to all regimes: haulage tariff $c_{\text{t}} = ₹6$ per tonne-kilometre; device mass $m = 0.19$ kg, giving $c_{\text{t}} \cdot m = ₹0.00114$ per device-kilometre.

Territorial regime	Effective road distance Δ (km)	Consignment fixed cost C_{cons} (₹)	Consolidation factor n	Fixed cost per device C_{cons}/n (₹)	Haulage cost per device $c_{\text{t}} \cdot \Delta \cdot m$ (₹)
Metropolitan core	25	400	200	2.00	0.03
Mid-periphery	180	900	40	22.50	0.21
Hinterland, no node	520	250	1	250.00	0.59
Hinterland with node	520	1400	60	23.33	0.59

Execution Feasibility Index components in Table 3 are assigned as follows. $A(r) = 1$ for refurbishment and material recycling, for which authorisation categories exist, and $A(r) = 0$ for component harvesting, for which none does. $\text{Acc}(d, r)$ is evaluated from the logistic form of Section 3.3 with the consolidation-adjusted effective distance. $\text{Cap}(r, t)$ is set at 0.80 in the metropolitan core and mid-periphery and 0.75 where dispatch is batched. $V(d, r) = 1$ where $S > 0$ and 0 otherwise, consistent with the minimum-across-chain definition. $L(d, r)$ reflects the share of the pathway passing through registered actors. Because A , V , and L are effectively binary at these parameter values, φ is driven almost entirely by whether an authorised executor exists and whether the pathway is value-positive, rather than by the continuous terms.

Appendix C. Computational artefacts

The following files reproduce Section 6.5.

Path	Role
analysis/aipeg_model.py	Equations (1)–(5)
analysis/run_experiment.py	Table 3 reproduction, 30-origin panel, Figures 1–5
data/published_indicators.csv	Cited national statistics with sources
data/district_origins.csv	Thirty Indian origins (lat/lon)
data/facility_hubs.csv	Nine processing-hub proxies
results/table3_reproduction.csv	Implemented Table 3
results/district_phi.csv	Origin–strategy panel
results/headline_tests.csv	T1–T3 summary
results/figures/	Figures 1–5

Run:

```
python3 analysis/run_experiment.py
```

Road distance is great-circle kilometres times 1.30. Hubs are proxies for a geocoded CPCB register that is not published as a research dataset. Appendix B rupee values are unchanged assumptions. The exercise tests the theory’s structure; it is not a national CER estimate.